RAOH

— Researchers —

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

Rayyan (Systematic Review Screening Tool)

Rayyan is a web-based platform designed to help researchers efficiently conduct **systematic reviews** and **meta-analyses**, specifically during the *screening phase* of studies.

In simple terms:

Rayyan is a tool used to organize and filter large numbers of scientific articles to decide which studies should be included in a review and which should be excluded.

Why Rayyan is Important

In a systematic review, researchers often retrieve a very large number of articles from databases such as:

- PubMed
- Scopus
- Web of Science
- Google Scholar

This can result in hundreds or even thousands of records.

Manually screening them is:

- Time-consuming
- Repetitive
- Prone to human error

Rayyan solves this problem by making the screening process:

- Faster
- More organized
- Less biased
- Collaborative

How Rayyan Works (Step-by-Step)

1. Import Studies

Researchers upload studies from databases into Rayyan.

At this stage, you may have:

- 500–5000 records

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2. De-duplication

Rayyan automatically helps identify and remove duplicate studies.

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3. Screening Phase (Core Function)

Each study is reviewed based on:

- Title
- Abstract
- Keywords

You classify each record into:

- **Include (✓)** → Relevant studies to be used in the review
- **Exclude (✗)** → Not relevant to the research question
- **Maybe (❓)** → Uncertain, needs further review

Example

Research topic:

Sleep quality among medical students

During screening:

- A study on nurses → ❌ Exclude
- A study on stress in medical students → ✓ Include
- A study on general university students → ❓ Maybe

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4. Tagging (Reason for Exclusion)

Rayyan allows you to label excluded studies with reasons such as:

- Wrong population
- Irrelevant outcome
- Wrong study design
- Duplicate
- Non-English article

This improves transparency in systematic reviews.

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5. Blinded Review (Reducing Bias)

One of Rayyan's strongest features is **blinded screening**.

This means:

- Reviewers cannot see each other's decisions initially
- This reduces selection bias
- Improves objectivity in study inclusion

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6. Collaboration

Multiple researchers can:

- Screen the same dataset
- Compare decisions
- Resolve disagreements

This is essential in academic systematic reviews.

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7. AI-Assisted Screening (Optional Feature)

Rayyan also provides AI-based suggestions that:

- Prioritize relevant studies
- Learn from your inclusion/exclusion patterns
- Speed up the screening process

Where Rayyan is Used

Rayyan is specifically used in:

- Systematic Reviews
- Meta-Analyses

It is **not used** for:

- Cross-sectional studies
- Cohort studies
- Randomized controlled trials (RCTs) as a design tool

Key Idea

Rayyan is not a data analysis tool — it is a **study selection and screening tool** used before analysis begins.

Simple Summary

Rayyan helps researchers to:

- Import thousands of articles
- Remove duplicates
- Screen titles and abstracts
- Decide inclusion/exclusion
- Collaborate with other reviewers
- Reduce bias in study selection